

Unit 5 Python Programs

```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

# Creating a common dataset
data = {
    "Name": ["Arun", "Bala", "Charan", "Divya", "Esha", "Farah", "Arun", "Hari"],
    "Age": [21, 22, np.nan, 24, 23, 22, 21, np.nan],
    "Department": ["ECE", "CSE", "ECE", "IT", "CSE", "ECE", "ECE", "IT"],
    "Marks": [85, 92, 78, np.nan, 88, 95, 85, 70],
    "Attendance": [90, 85, 78, 92, np.nan, 88, 90, 75]
}

df = pd.DataFrame(data)

print(df)
```

	Name	Age	Department	Marks	Attendance
0	Arun	21.0	ECE	85.0	90.0
1	Bala	22.0	CSE	92.0	85.0
2	Charan	NaN	ECE	78.0	78.0
3	Divya	24.0	IT	NaN	92.0
4	Esha	23.0	CSE	88.0	NaN
5	Farah	22.0	ECE	95.0	88.0
6	Arun	21.0	ECE	85.0	90.0
7	Hari	NaN	IT	70.0	75.0

1. Filtering data

```
filtered_data = df[df["Marks"] > 85]

print(filtered_data)
```

	Name	Age	Department	Marks	Attendance
1	Bala	22.0	CSE	92.0	85.0
4	Esha	23.0	CSE	88.0	NaN
5	Farah	22.0	ECE	95.0	88.0

2. Filtering Data with Multiple Conditions

```
filtered_data = df[
    (df["Marks"] > 80) &
    (df["Attendance"] > 85)
]

print(filtered_data)
```

	Name	Age	Department	Marks	Attendance
0	Arun	21.0	ECE	85.0	90.0
5	Farah	22.0	ECE	95.0	88.0
6	Arun	21.0	ECE	85.0	90.0

3. Sorting Data in Both Orders

Ascending Order

```
ascending_data = df.sort_values(by="Marks", ascending=True)

print(ascending_data)
```

	Name	Age	Department	Marks	Attendance
7	Hari	NaN	IT	70.0	75.0
2	Charan	NaN	ECE	78.0	78.0
6	Arun	21.0	ECE	85.0	90.0
0	Arun	21.0	ECE	85.0	90.0
4	Esha	23.0	CSE	88.0	NaN
1	Bala	22.0	CSE	92.0	85.0
5	Farah	22.0	ECE	95.0	88.0
3	Divya	24.0	IT	NaN	92.0

Descending Order

```
descending_data = df.sort_values(by="Marks", ascending=False)

print(descending_data)
```

	Name	Age	Department	Marks	Attendance
5	Farah	22.0	ECE	95.0	88.0
1	Bala	22.0	CSE	92.0	85.0
4	Esha	23.0	CSE	88.0	NaN
6	Arun	21.0	ECE	85.0	90.0
0	Arun	21.0	ECE	85.0	90.0
2	Charan	NaN	ECE	78.0	78.0
7	Hari	NaN	IT	70.0	75.0
3	Divya	24.0	IT	NaN	92.0

4. Handling Missing Values

Detecting Missing Values

```
missing_values = df.isnull()

print(missing_values)
```

	Name	Age	Department	Marks	Attendance
0	False	False	False	False	False
1	False	False	False	False	False
2	False	True	False	False	False
3	False	False	False	True	False
4	False	False	False	False	True
5	False	False	False	False	False
6	False	False	False	False	False
7	False	True	False	False	False

Counting Missing Values

```
missing_count = df.isnull().sum()

print(missing_count)
```

```
Name      0
Age        2
Department 0
Marks      1
Attendance 1
dtype: int64
```

5. Removing Missing Values

```
clean_df = df.dropna()

print(clean_df)
```

	Name	Age	Department	Marks	Attendance
0	Arun	21.0	ECE	85.0	90.0
1	Bala	22.0	CSE	92.0	85.0
5	Farah	22.0	ECE	95.0	88.0
6	Arun	21.0	ECE	85.0	90.0

6. Removing Duplicate Data

```
unique_df = df.drop_duplicates()

print(unique_df)
```

	Name	Age	Department	Marks	Attendance
0	Arun	21.0	ECE	85.0	90.0
1	Bala	22.0	CSE	92.0	85.0
2	Charan	NaN	ECE	78.0	78.0
3	Divya	24.0	IT	NaN	92.0
4	Esha	23.0	CSE	88.0	NaN
5	Farah	22.0	ECE	95.0	88.0
7	Hari	NaN	IT	70.0	75.0

7. Aggregating the Data

```
aggregation = df.groupby("Department").agg({
    "Marks": "mean",
    "Attendance": "mean",
    "Age": "mean"
})

print(aggregation)
```

Department	Marks	Attendance	Age
CSE	90.00	85.0	22.500000
ECE	85.75	86.5	21.333333
IT	70.00	83.5	24.000000

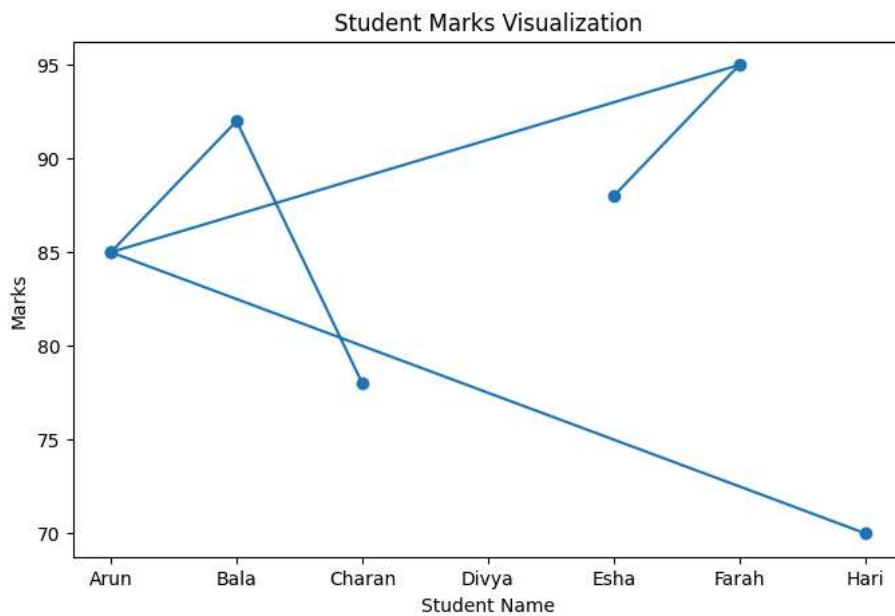
8. Data Visualization (Line Plot)

```
plt.figure(figsize=(8, 5))

plt.plot(df["Name"], df["Marks"], marker="o")

plt.xlabel("Student Name")
plt.ylabel("Marks")
plt.title("Student Marks Visualization")

plt.show()
```



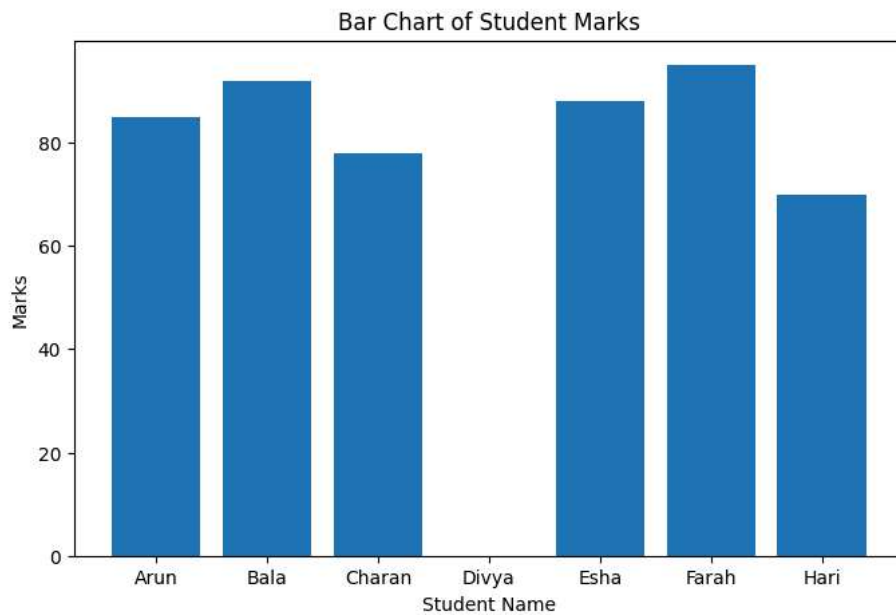
9. Bar Chart

```
plt.figure(figsize=(8, 5))

plt.bar(
    df["Name"],
    df["Marks"]
)

plt.xlabel("Student Name")
plt.ylabel("Marks")
plt.title("Bar Chart of Student Marks")

plt.show()
```



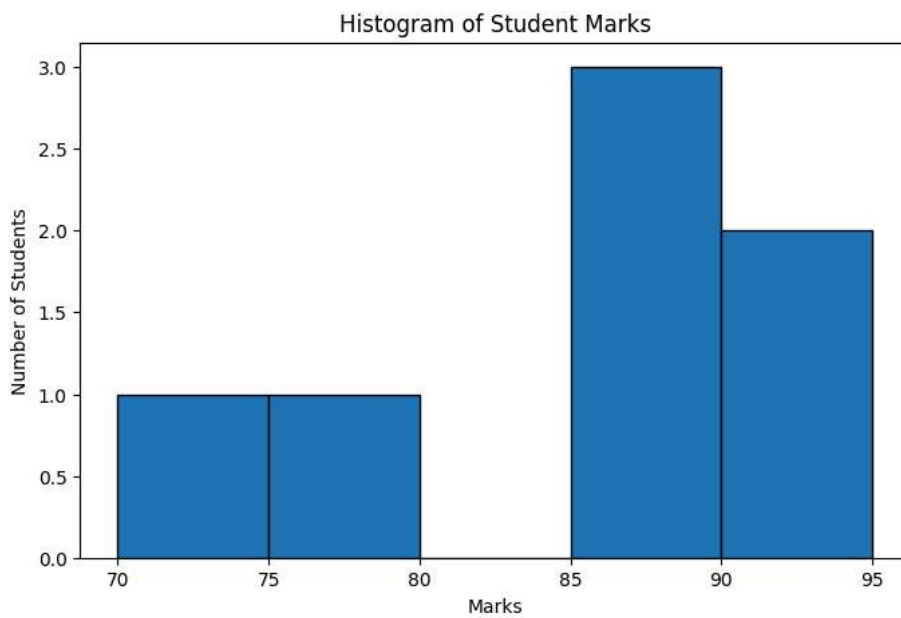
10. Histogram

```
plt.figure(figsize=(8, 5))

plt.hist(
    df["Marks"].dropna(),
    bins=5,
    edgecolor="black"
)

plt.xlabel("Marks")
plt.ylabel("Number of Students")
plt.title("Histogram of Student Marks")

plt.show()
```



11. Scatter Plot

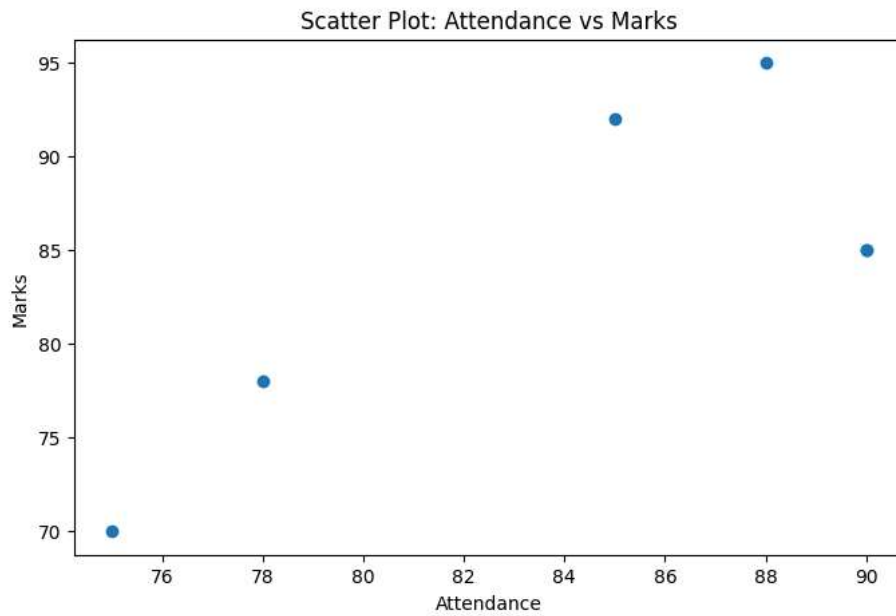
```
plot_df = df.dropna(subset=["Attendance", "Marks"])

plt.figure(figsize=(8, 5))
```

```
plt.scatter(
    plot_df["Attendance"],
    plot_df["Marks"]
)

plt.xlabel("Attendance")
plt.ylabel("Marks")
plt.title("Scatter Plot: Attendance vs Marks")

plt.show()
```



12. Pie Chart

```
department_count = df["Department"].value_counts()

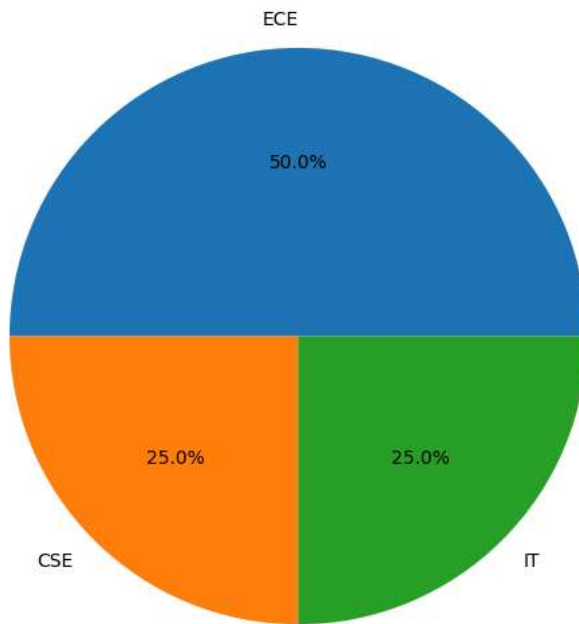
plt.figure(figsize=(7, 7))

plt.pie(
    department_count,
    labels=department_count.index,
    autopct="%1.1f%%"
)

plt.title("Department-wise Student Distribution")

plt.show()
```

Department-wise Student Distribution



13. Plot Labels and Titles

```
plot_df = df.dropna(subset=["Marks"])

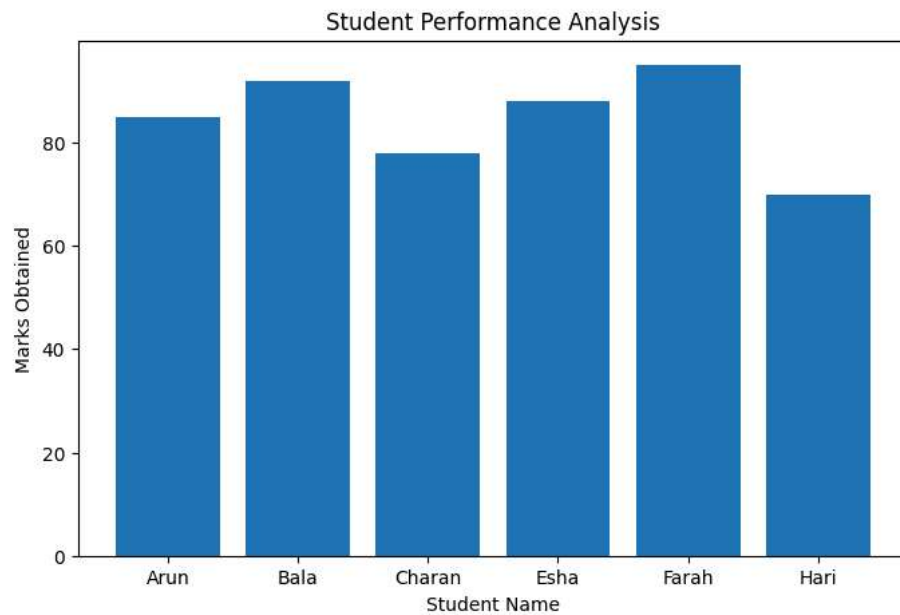
plt.figure(figsize=(8, 5))

plt.bar(
    plot_df["Name"],
    plot_df["Marks"]
)

# Plot Labels
plt.xlabel("Student Name")
plt.ylabel("Marks Obtained")

# Plot Title
plt.title("Student Performance Analysis")

plt.show()
```



14. Subplots

```
plot_df = df.dropna(subset=["Marks", "Attendance"])
```

```
plt.figure(figsize=(12, 8))
```

```
# Subplot 1 - Line Plot
```

```
plt.subplot(2, 2, 1)
```

```
plt.plot(  
    plot_df["Name"],  
    plot_df["Marks"],  
    marker="o"  
)
```

```
plt.title("Line Plot")
```

```
plt.xlabel("Name")
```

```
plt.ylabel("Marks")
```

```
# Subplot 2 - Bar Chart
```

```
plt.subplot(2, 2, 2)
```

```
plt.bar(  
    plot_df["Name"],  
    plot_df["Marks"]  
)
```

```
plt.title("Bar Chart")
```

```
plt.xlabel("Name")
```

```
plt.ylabel("Marks")
```

```
# Subplot 3 - Histogram
```

```
plt.subplot(2, 2, 3)
```

```
plt.hist(  
    plot_df["Marks"],  
    bins=5,  
    edgecolor="black"  
)
```

```
plt.title("Histogram")
```

```
plt.xlabel("Marks")
```

```
plt.ylabel("Frequency")
```

```
# Subplot 4 - Scatter Plot
```

```
plt.subplot(2, 2, 4)
```

```
plt.scatter(  
    plot_df["Attendance"],  
    plot_df["Marks"]  
)
```

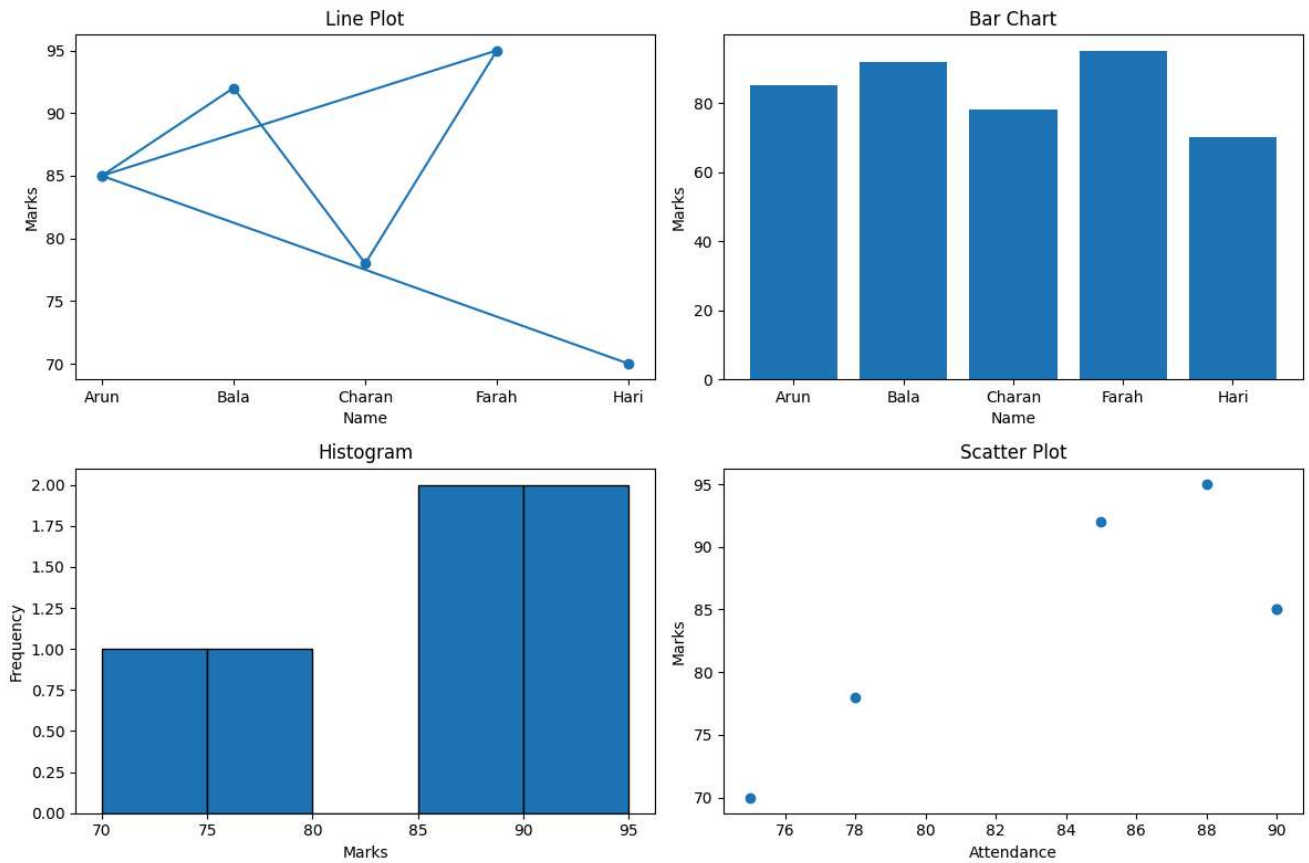
```
plt.title("Scatter Plot")
```

```
plt.xlabel("Attendance")
```

```
plt.ylabel("Marks")
```

```
plt.tight_layout()
```

```
plt.show()
```



15. Reading and Writing CSV File

Writing the DataFrame to a CSV File

```
df.to_csv("student_data.csv", index=False)
```

```
print("CSV file created successfully!")
```

```
CSV file created successfully!
```

Reading the CSV File

```
new_df = pd.read_csv("student_data.csv")
```

```
print(new_df)
```

	Name	Age	Department	Marks	Attendance
0	Arun	21.0	ECE	85.0	90.0
1	Bala	22.0	CSE	92.0	85.0
2	Charan	NaN	ECE	78.0	78.0
3	Divya	24.0	IT	NaN	92.0
4	Esha	23.0	CSE	88.0	NaN
5	Farah	22.0	ECE	95.0	88.0
6	Arun	21.0	ECE	85.0	90.0
7	Hari	NaN	IT	70.0	75.0

Head()

```
print(df.head())
print(df.head(3))
```

	Name	Age	Department	Marks	Attendance
0	Arun	21.0	ECE	85.0	90.0
1	Bala	22.0	CSE	92.0	85.0
2	Charan	NaN	ECE	78.0	78.0
3	Divya	24.0	IT	NaN	92.0
4	Esha	23.0	CSE	88.0	NaN

	Name	Age	Department	Marks	Attendance
0	Arun	21.0	ECE	85.0	90.0
1	Bala	22.0	CSE	92.0	85.0
2	Charan	NaN	ECE	78.0	78.0

Tail()

```
print(df.tail())
print(df.tail(3))
```

	Name	Age	Department	Marks	Attendance
3	Divya	24.0	IT	NaN	92.0
4	Esha	23.0	CSE	88.0	NaN
5	Farah	22.0	ECE	95.0	88.0
6	Arun	21.0	ECE	85.0	90.0
7	Hari	NaN	IT	70.0	75.0

	Name	Age	Department	Marks	Attendance
5	Farah	22.0	ECE	95.0	88.0
6	Arun	21.0	ECE	85.0	90.0
7	Hari	NaN	IT	70.0	75.0

Info()

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8 entries, 0 to 7
Data columns (total 5 columns):
#   Column      Non-Null Count  Dtype
---  -
0   Name        8 non-null     object
```